TWO-COURSE PILOT | DISCUSSION DRAFT

Reliable computing for teaching

What we have previously relied on can no longer be guaranteed

Request managed CPU CloudBank Classroom access through ACCESS for a data science + MLIS pilot. Fund the local service around it, prove one assignment per course, then admit an 80-seat semester only after the readiness checks pass.

Planning baseline: 40 data science students + 40 MLIS students; six preparation weeks, 15 teaching weeks and two export/close weeks. Actual courses, term, instructors, PI, resource limits and operating commitments are pending.

Decision requested

Approve a preparation/canary ceiling of $6,000 within an illustrative total cash ceiling of $30,000. Release the remaining envelope only after the pre-class checks have dated evidence. Reserve committed faculty/Libraries/finance time separately. This is a proposed staged cap, not an existing authorization.

One component before a semester

8–6 weeks before term - Authorize the plan

Choose actual course titles/dates and owners. Approve up to $6,000 for preparation/canary within an illustrative $30,000 total ceiling. Resolve institutional data/support requirements and confirm CloudBank's route before the designated PI submits.

6–4 weeks before term - Run a ten-person component canary

After approved access, recruit up to ten consenting volunteers, five per course concept. Run one DS notebook and one metadata/database/export workflow with synthetic inputs. Measure setup, package/memory fit, save/restart/export and support effort. No real grades or private library data are needed.

4–2 weeks before term - Rehearse the whole class and recovery

With provider approval, rehearse 80 student workspaces plus agreed staff use. Test account isolation, grading receipts, accessible instructions, independent restore and the CPU fallback. Update the cost/usage forecast from observed operator-hours.

Weeks 1–15 - Teach the two-course pilot

Publish orientation and support windows. Keep all required assessment CPU-capable. Review access, usage, storage, tickets and backups weekly; brief course leads at weeks 3, 8 and 12. Optional GPU, OSPool and MESA modules have separate admission checks.

Term end + two weeks - Export, close and decide

Provide learner exports and reconcile grades before closure. Preserve approved public outputs with suitable metadata; apply the institutional retention/deletion schedule to private records. Do not assume educational allocation extension. Decide whether next term needs a new project or confirmed renewal.

ASSIGNMENTS AND STAFFING

Two CPU workflows; named support

INFO 521: Introduction to Machine Learning - 40 planned seats

Build a reproducible data-cleaning and exploratory-analysis notebook, then fit a small CPU model with held-out validation. Explain missingness, leakage, uncertainty and data provenance.

Inputs: Generate a synthetic civic-service dataset with no real residents; keep core CSV inputs below 250 MB and test peak memory against the awarded workspace. Real/public datasets require a rights and data-classification check.

Environment: Requested: Python, JupyterLab, pandas, scikit-learn, matplotlib and an approved reproducible package manifest; R/RStudio only if the course requires it.

Acceptance: Run the whole assignment from a fresh student workspace; require execution without out-of-memory errors at the approved spec, reproducible key outputs and a complete exported submission.

LIS 671: Digital Curation and Digital Preservation - 40 planned seats

Audit a collection catalog, normalize names/dates/identifiers, document a metadata crosswalk, query an embedded catalog database and export a preservation-ready package with README, schema, provenance and checksums.

Inputs: Use a synthetic 10,000–100,000-record collection catalog with Dublin Core-style fields. Add public/licensed records only after rights review; exclude patron, circulation, donor and restricted collection records.

Environment: Requested: JupyterLab/Python, CSV/JSON/XML parsing and embedded SQLite; optional MARC parsing packages after approval. OpenRefine or other hosted applications require a separate provider fit check.

Acceptance: Execute the complete workflow in a fresh workspace; validate schema, record counts, identifiers and query results; reopen the exported package using the fallback environment.

Staffing coverage

Primary operator: four hours per teaching week, with 40 preparation and 12 close hours; backup: one hour/week, eight prep and four close. Two TAs: three hours per course/week plus four prep and two close hours each. Each course lead reserves 35 additional pilot hours. Normal course instruction/grading is outside this added-effort estimate.

Publish staffed help windows and a proposed one-business-day first-response target; no 24/7 promise. Escalation: TA/course lead to primary operator, backup/provider and service owner. Primary and backup operators must be different people.

GPU, MESA/CyVerse, NRP and OSPool modules need separate gates. PATh/IDSS eligibility is unconfirmed. Optional resources cannot become required assessment dependencies without their own checks.

ILLUSTRATIVE USD COSTS - NOT PAY SCALES OR QUOTES

Fund the operations, not only the resource

Role	Hours	Loaded rate	Mode	Cost
Service owner / allocation lead	16	$100	cash	$1,600
Primary technical operator	112	$90	cash	$10,080
Backup technical operator	27	$90	cash	$2,430
Two course support TAs	102	$35	cash	$3,570
Instructional design / accessibility	24	$75	cash	$1,800
IT security / privacy reviewer	12	$100	cash	$1,200
Two course leads	70	$90	in-kind	$6,300
Libraries / data steward	16	$85	in-kind	$1,360
College sponsor / finance	8	$70	in-kind	$560

Approved backup/export destinations: $2,000 reserve.

Durable exhibit / database destination: $600 reserve.

Fallback / unquoted service costs: $1,500 reserve.

Cash base $24,780 + 20% contingency $4,956 = $29,736. Required in-kind effort $8,220. Cash + committed-time value $37,956.

Illustrative USD planning rates, not UArizona pay scales, provider quotes or approved funds. Rates are assumed fully loaded (salary/fringe/overhead included); do not add fringe twice. In-kind effort is a required written workload commitment, not free labor. Replacing defaults changes the proposal only, not an approved budget.

Assume no local compute payment only if the awarded Classroom resource covers the approved use. Confirm actual units, credit exchange, included storage/support and any uncovered charges; $0 is not a provider quote. The Jetstream2 SU model must not be entered as CloudBank credits.

Excludes normal course instruction/grading, new staff recruitment, major platform development, laptops, paid software, optional GPU/model charges and institutional indirect costs beyond the assumed loaded rates. Additional scope requires a new estimate. No student-paid account or personal payment card is proposed.

Editable budget and conditional campus-storage comparison

TWELVE CHECKS - ALL PLANNED / NOT RUN

No classroom admission without evidence

1. Scope | Sponsor + service owner

Named eligible PI, leads, distinct operators and committed funds/coverage.

2. Provider | Service owner + CloudBank

Approved Classroom route, 80 student + six staff scope, actual limits and term/accounting.

3. Accounts | Operator + course leads

Approved logins; student/instructor isolation and revocation demonstrated.

4. Software | Each course lead

Both full CPU assignments run cleanly on the pinned environment.

5. Capacity | Operator + provider

Provider-approved class-load rehearsal; proposed p95 workspace access within five minutes.

6. Save | Course leads + TAs

Save, stop/restart, export and reopen in the CPU fallback.

7. Grades | Course leads

Versioned submissions and complete test-grade receipts reconcile in the approved LMS.

8. Data | IT/privacy reviewer + Libraries

Approved data destinations, rights, access roles and retention; no private records on the public site.

9. Accessibility | Instructional designer + course leads

Essential keyboard/screen-reader workflows and equivalent assessed CPU/offline route.

10. Recovery | Primary + backup operators

Restore on another environment; measured proposed RPO 24 hours/RTO one business day.

11. Support | Owner + TAs + operators

Published staffed help/escalation and a rehearsed outage fallback; no 24/7 promise.

12. Exit | Service owner + course leads + Libraries

Expiry alerts, verified learner/grade exports, closure/deletion and confirmed next-term decision.

Evidence, evaluation and exit

Keep private account/grade evidence in the institutional register; publish aggregate results only. Measure access and clean-run assignment success, exports/grade receipts, operator-hours, support demand, actual charges and independent-restore timing. Before expiry, provide learner exports, reconcile grades and confirm next-term or closure/retention/deletion arrangements.

Full readiness register and evidence criteria

Cohesive program site and source library